

Height-Resolved Curvature Decomposition: Separating Real Stability Curvature from Coordinate Effects

The 2D contour diagnostic panel demonstrates that sharp vertical profile features near the critical Richardson number ($Ri_c \approx 0.20$) often stem from coordinate stretching rather than purely local thermodynamic phase changes. When turbulent flux varies with height, the local Obukhov length scale $L(z)$ distorts the vertical coordinate frame.

We decompose the total profile curvature of $Ri(z)$ into distinct physical and numerical components:

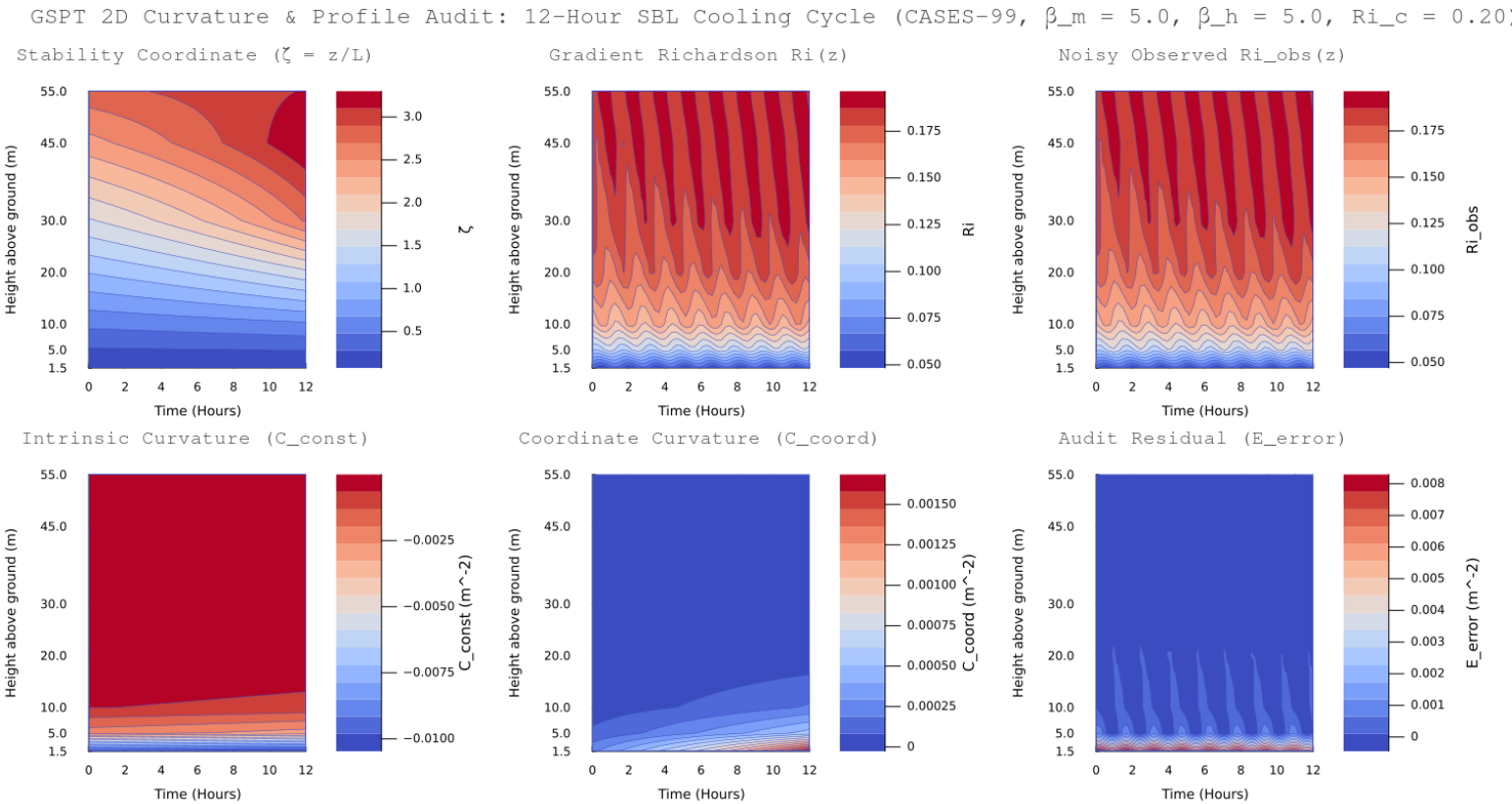
$$Ri_{zz} = \underbrace{Ri_{\zeta\zeta}\zeta_z^2}_{C_{\text{const}}} + \underbrace{Ri_{\zeta}\zeta_{zz}}_{C_{\text{coord}}} + E_{\text{error}}$$

- **Intrinsic Stability Curvature (C_{const}):** Expected curvature from local thermodynamic stability holding the similarity coordinate fixed. Under Monin–Obukhov similarity ($\beta_m = 5.0, \beta_h = 5.0$), this term remains negative, reflecting gradual lower-level stabilization.
- **Coordinate-Stretching Curvature (C_{coord}):** Curvature introduced because $\zeta(z) = z/L(z)$ varies with height under flux-divergent SBL conditions. Near nocturnal low-level jet (LLJ) noses, this coordinate effect spikes strongly positive, counteracting intrinsic cooling.
- **Audit Residual (E_{error}):** Quantifies operator truncation error ($\mathbf{D}_2 Ri_{\text{smooth}} - Ri_{\text{exact}}$). Keeping numerical errors separate prevents finite-difference artifacts on non-uniform CASES-99 tower grids (1.5 m to 55 m) from contaminating physical curvature fields.

Key Physical Insights (12-Hour SBL Cooling Cycle)

- **Near-Surface Cancellation ($z \approx 10$ m):** During deep stable hours (Hours 6–12), positive coordinate curvature ($C_{\text{coord}} \approx +0.0036 \text{ m}^{-2}$) cancels negative stability curvature ($C_{\text{const}} \approx -0.0040 \text{ m}^{-2}$). The resulting locally linear profile ($Ri_{zz} \approx -0.0004 \text{ m}^{-2}$) hides active, offsetting physical processes.
- **Noise Suppression & Filtering:** Synthetic sensor jitter ($\sigma_{\text{obs}} = 0.008$) added to Ri_{obs} is smoothed via Tikhonov regularized differentiation ($\lambda_{\text{reg}} = 5.0$) before operator evaluation,

isolating physical signal from observational noise.



6-Panel Diagnostic Layout

Panel Position	Variable	Physical / Numerical Interpretation
Top Left	Stability Coordinate (ζ)	Dimensionless height ratio $z/L(z)$ across time and altitude.
Top Center	Gradient Richardson (Ri)	Exact analytical profile derived from SBL similarity theory.
Top Right	Observed Richardson (Ri_{obs})	Synthetic sensor data containing Gaussian observational jitter.
Bottom Left	Intrinsic Curvature (C_{const})	Pure thermodynamic curvature assuming constant coordinate scaling.
Bottom	Coordinate Curvature	Curvature generated by vertical flux divergence and

Center	(C_{coord})	LLJ shear.
Bottom Right	Audit Residual (E_{error})	Explicit finite-difference discretization residual ($\mathbf{D}_2 Ri_{\text{smooth}} - Ri_{\text{exact}}$).

```
#!/usr/bin/env julia
# src/gspt_curvature_contour_v3.jl
# version 3 of the GSPT curvature contour plotting script
# David E. England, Ph.D.
# email: david.england@uah.edu
using LinearAlgebra
using Printf

# Try to use Plots.jl
try
    using Plots
catch e
    @warn "Plots.jl package not found in current environment. Script will save numerical data to file."
end

function stencil_weights(z_stencil::Vector{Float64}, z0::Float64, m::Int)
    p = length(z_stencil)
    A = [Float64(z - z0)^(k - 1) / factorial(k - 1) for k in 1:p, z in z_stencil]
    b = zeros(p)
    b[m + 1] = 1.0 # Select target derivative order
    return A \ b
end

function build_operators(z::Vector{Float64})
    n = length(z)
    D1 = zeros(n, n)
    D2 = zeros(n, n)

    for i in 1:n
        if i == 1
            idx = [1, 2, 3]
        elseif i == n
            idx = [n-2, n-1, n]
        else
            idx = [i-2, i-1, i, i+1, i+2]
        end
        # Compute first and second derivatives
        for j in 1:n
            if i != j
                D1[i, j] = (z[j] - z[i])^-1
                D2[i, j] = (z[j] - z[i])^-2
            end
        end
    end
end
```

```

        else
            idx = [i-1, i, i+1]
        end

        D1[i, idx] = stencil_weights(z[idx], z[i], 1)
        D2[i, idx] = stencil_weights(z[idx], z[i], 2)
    end

    return D1, D2
end

Ri_model( $\zeta$ ,  $\beta_m$ ,  $\beta_h$ ) =  $\zeta * (1.0 + \beta_h * \zeta) / (1.0 + \beta_m * \zeta)^2$ 

function Ri_zeta( $\zeta$ ,  $\beta_m$ ,  $\beta_h$ )
    return  $(1.0 + (2.0 * \beta_h - \beta_m) * \zeta) / (1.0 + \beta_m * \zeta)^3$ 
end

function Ri_zetazeta( $\zeta$ ,  $\beta_m$ ,  $\beta_h$ )
    num =  $2.0 * \beta_h - 4.0 * \beta_m + 2.0 * \beta_m * (\beta_m - 2.0 * \beta_h) * \zeta$ 
    return num /  $(1.0 + \beta_m * \zeta)^4$ 
end

function plot_gspt_curvature_layers(
    hours::Vector{Float64},
    z_tower::Vector{Float64},
    zeta_mat::Matrix{Float64},
    ri_smooth_mat::Matrix{Float64},
    ri_obs_mat::Matrix{Float64},
    C_const::Matrix{Float64},
    C_coord::Matrix{Float64},
    E_error::Matrix{Float64};
    save_path::String = "gspt_curvature_contour_v3.png"
)
    if !@isdefined Plots
        @warn "Plots.jl not loaded. Skipping PNG rendering."
        return nothing
    end

    gr() # Set GR backend

    # Common plotting parameters

```

```
plt_opts = (
    xlabel = "Time (Hours)",
    ylabel = "Height above ground (m)",
    yticks = (z_tower, [sprintf("%.1f", h) for h in z_tower]),
    fill = true,
    c = :coolwarm,
    linewidth = 0.5,
    tickfont = font(8, "sans-serif"),
    guidefont = font(9, "sans-serif"),
    titlefont = font(10, "sans-serif bold")
)
```

```
# Row 1 plots
```

```
p1 = contourf(hours, z_tower, zeta_mat;
    title="Stability Coordinate ( $\zeta = z/L$ )",
    colorbar_title=" $\zeta$ ",
    plt_opts...)
```

```
p2 = contourf(hours, z_tower, ri_smooth_mat;
    title="Gradient Richardson  $Ri(z)$ ",
    colorbar_title="Ri",
    plt_opts...)
```

```
p3 = contourf(hours, z_tower, ri_obs_mat;
    title="Noisy Observed  $Ri_{obs}(z)$ ",
    colorbar_title="Ri_obs",
    plt_opts...)
```

```
# Row 2 plots
```

```
p4 = contourf(hours, z_tower, C_const;
    title="Intrinsic Curvature ( $C_{const}$ )",
    colorbar_title="C_const ( $m^{-2}$ )",
    plt_opts...)
```

```
p5 = contourf(hours, z_tower, C_coord;
    title="Coordinate Curvature ( $C_{coord}$ )",
    colorbar_title="C_coord ( $m^{-2}$ )",
    plt_opts...)
```

```
p6 = contourf(hours, z_tower, E_error;
    title="Audit Residual ( $E_{error}$ )",
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        colorbar_title="E_error (m-2)",
        plt_opts...)

# Assemble into a 2x3 diagnostic panel layout
full_plot = plot(p1, p2, p3, p4, p5, p6,
                 layout = (2, 3),
                 size = (1500, 800),
                 plot_title = "GSPT 2D Curvature & Profile Audit: 12-Hour SBL Cool:",
                 plot_titlefont = font(12, "sans-serif bold"),
                 margin = 6Plots.mm)

savefig(full_plot, save_path)
println("Successfully rendered and saved: $save_path")
return full_plot
end

function run_test_simulation()
    println("Setting up SCM 12-hour cooling cycle...")

    # CASES-99 55m Tower heights
    z_tower = [1.5, 5.0, 10.0, 20.0, 30.0, 45.0, 55.0]
    n_z = length(z_tower)
    D1, D2 = build_operators(z_tower)

    # Time steps: 50 points over 12 hours
    hours = collect(range(0.0, 12.0, length=50))
    n_t = length(hours)

    # Curvature & profile matrices
    zeta_mat = zeros(n_z, n_t)
    ri_smooth_mat = zeros(n_z, n_t)
    ri_obs_mat = zeros(n_z, n_t)
    C_const_mat = zeros(n_z, n_t)
    C_coord_mat = zeros(n_z, n_t)
    E_error_mat = zeros(n_z, n_t)

    # SBL model settings (using user-requested grassland parameters)
    β_m, β_h = 5.0, 5.0
    L0 = 20.0
    sigma_obs = 0.008
    lambda_reg = 5.0

```

```

for t_idx in 1:n_t
    t = hours[t_idx]

    # Non-stationary LLJ Jet nose intensity growing with cooling
    amp = 0.1 + 0.35 * (t / 12.0)
    L_profile = [L0 * (1.0 - amp * sin( $\pi$  * zi / 60.0)) for zi in z_tower]

    # Similarity coordinate profile
     $\zeta$  = z_tower ./ L_profile
     $\zeta_z$  = D1 *  $\zeta$ 
     $\zeta_{zz}$  = D2 *  $\zeta$ 

    zeta_mat[:, t_idx] =  $\zeta$ 

    # Analytical GSPT curvature
    ri_z = [Ri_zeta( $\zeta$ [i],  $\beta_m$ ,  $\beta_h$ ) for i in 1:n_z]
    ri_zz = [Ri_zetazeta( $\zeta$ [i],  $\beta_m$ ,  $\beta_h$ ) for i in 1:n_z]

    C_const = ri_zz .* ( $\zeta_z$  .^ 2)
    C_coord = ri_z .*  $\zeta_{zz}$ 
    Ri_exact = C_const .+ C_coord

    C_const_mat[:, t_idx] = C_const
    C_coord_mat[:, t_idx] = C_coord

    # Add random sensor jitter (using deterministic seed offsets for repeatability)
    noise = sigma_obs .* sin.(t_idx .+ (1:n_z) .* 1.5)
    Ri_obs = [Ri_model( $\zeta$ [i],  $\beta_m$ ,  $\beta_h$ ) for i in 1:n_z] .+ noise
    ri_obs_mat[:, t_idx] = Ri_obs

    # Track A: Tikhonov regularization filter
    R = D2' * D2
    A_reg = I(n_z) + lambda_reg .* R
    Ri_smooth = A_reg \ Ri_obs
    ri_smooth_mat[:, t_idx] = Ri_smooth

    # Computed physical curvature
    M_Ri_zz = D2 * Ri_smooth

    # Discrete audit error

```

```
        E_error_mat[:, t_idx] = M_Ri_zz .- Ri_exact
    end

    println("Matrices populated successfully. Generating Plots panel...")

    plot_gspt_curvature_layers(hours, z_tower, zeta_mat, ri_smooth_mat, ri_obs_mat,
                               C_const_mat, C_coord_mat, E_error_mat;
                               save_path = "gspt_curvature_contour_v3.png")
end

run_test_simulation()
```